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Method and apparatus for spinal facet fusion

Abstract

A spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another.

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Background/Summary

REFERENCE TO PENDING PRIOR PATENT APPLICATIONS (1) This patent application is a continuation of pending prior U.S. patent application Ser. No. 16/879,385, filed May 20, 2020 by VGI Medical, LLC and Tov Vestgaarden et al. for METHOD AND APPARATUS FOR SPINAL FACET FUSION, which patent application is a continuation of prior U.S. patent application Ser. No. 15/631,670, filed Jun. 23, 2017 by VGI Medical, LLC for METHOD AND APPARATUS FOR SPINAL FACET FUSION, which patent application claims benefit of prior U.S. Provisional Patent Application Ser. No. 62/353,809, filed Jun. 23, 2016 by VGI Medical, LLC and Tov Vestgaarden et al. for METHOD AND APPARATUS FOR SPINAL FACET FUSION. (2) The three (3) above-identified patent applications are hereby incorporated herein by reference.

FIELD OF THE INVENTION

(1) This invention relates to surgical methods and apparatus in general, and more particularly to surgical methods and apparatus for fusing spinal facets.

BACKGROUND OF THE INVENTION

- (2) Disc herniation is a condition where a spinal disc bulges from between two vertebral bodies and impinges on adjacent nerves, thereby causing pain. The current standard of care for surgically treating disc herniation in patients who have chronic pain and who have (or are likely to develop) associated spinal instability is spinal fixation. Spinal fixation procedures are intended to relieve the impingement on the nerves by removing the portion of the disc and/or bone responsible for compressing the neural structures and destabilizing the spine. The excised disc or bone is replaced with one or more intervertebral implants, or spacers, placed between the adjacent vertebral bodies.
- (3) In some cases, the spinal fixation leaves the affected spinal segment unstable. In this case, the spinal facets (i.e., the bony fins extending upwardly and downwardly from the rear of each vertebral body) can misengage with one another. The misengagement of the spinal facets can result in substantial pain to the patient. Furthermore, when left untreated, such misengagement of the spinal facets can result in the degeneration of the cartilage located between opposing facet surfaces, ultimately resulting in osteoarthritis, which can in turn lead to worsening pain for the patient.
- (4) Thus, where the patient suffers from spinal instability, it can be helpful to stabilize the facet joints as well as the vertebral bodies. The facet joints are frequently stabilized by fusing the spinal facets in position relative to one another.
- (5) In addition to providing stability, fusing the spinal facets can also be beneficial in other situations as well. By way of example but not limitation, osteoarthritis (a condition involving the degeneration, or wearing away, of the cartilage at the end of bones) frequently occurs in the facet joints. The prescribed treatment for osteoarthritis disorders depends on the location, severity and duration of the disorder. In some cases, non-operative procedures (including bed rest, medication, lifestyle modifications, exercise, physical therapy, chiropractic care and steroid injections) may be satisfactory treatment. However, in other cases, surgical intervention may be necessary. In cases where surgical intervention is prescribed, spinal facet fusion may be desirable.
- (6) A minimally-invasive, percutaneous approach for fusing spinal facets was proposed by Stein et al. ("Stein") in 1993. The Stein approach involved using a conical plug, made from cortical bone and disposed in a hole formed intermediate the spinal facet joint, to facilitate the fusing of opposing facet surfaces. However, the clinical success of this approach was limited. This is believed to be because the Stein approach did not adequately restrict facet motion. In particular, it is believed that movement of Stein's conical plug within its hole permitted unwanted facet movement to occur, thereby undermining facet fusion. Furthermore, the Stein approach also suffered from plug failure and plug migration.
- (7) Thus there is a need for a new and improved approach for effecting spinal facet fusion.
- (8) In addition to the foregoing, it should be appreciated that the spine comprises various regions having differing characteristics. More particularly, the first seven vertebrae (C1-C7) are the so-called cervical vertebrae, the next twelve vertebrae (T1-T12) are the so-called thoracic vertebrae, and the next five vertebrae (L1-L5) are the so-called lumbar vertebrae. Beneath the lumbar vertebrae are the five fused vertebrae of the sacrum, and then the four fused vertebrae of the coccyx (or tailbone). The facet joints in the cervical vertebrae can differ somewhat from the facet joints in the thoracic vertebrae and lumbar vertebrae, e.g., the facet joints in the cervical vertebrae are typically oriented generally horizontal to the longitudinal axis of the spine, whereas the facet joints of the thoracic vertebrae and the facet joints of the lumbar vertebrae are typically oriented generally vertical to the longitudinal axis of the spine. It has been found that it can be significantly more difficult to successfully effect spinal facet fusion in the cervical vertebrae than in the thoracic vertebrae and in the lumbar vertebrae.
- (9) Thus there is a need for a new and improved approach for effecting spinal facet fusion in the cervical vertebrae.

SUMMARY OF THE INVENTION

- (10) The present invention provides a novel method and apparatus for effecting spinal facet fusion, and is particularly advantageous for use in effecting spinal facet fusion in the cervical vertebrae.

More particularly, the present invention comprises the provision and use of a novel spinal facet fusion implant (sometimes hereinafter referred to as a “novel fusion implant”) for disposition between the opposing articular surfaces of a facet joint, including the facet joint of a cervical vertebrae, whereby to immobilize the facet joint and facilitate fusion between the opposing facets. The present invention also comprises the provision and use of novel instrumentation for installing the novel spinal facet fusion implant in a facet joint.

(11) In one preferred form of the invention, there is provided a spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another.

(12) In another preferred form of the invention, there is provided a system for effecting spinal facet fusion, the system comprising: a spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another; a drill guide/cannula for preparing the anatomy to receive the spinal facet fusion implant and for delivering the spinal facet fusion implant to the anatomy, the drill guide/cannula comprising: a body having a distal end, a proximal end and a longitudinal axis extending therebetween, the distal end of the body being configured for engaging the gap between a descending facet of a first vertebra and an ascending facet of a second vertebra; a central lumen extending between the distal end of the body and the proximal end of the body, the central lumen having a cross-sectional profile which matches the cross-sectional profile of the spinal facet fusion implant such that the spinal facet fusion implant can be introduced into the proximal end of the central lumen, advanced distally along the central lumen, and advanced distally out of the distal end of the central lumen and into the gap between the descending facet of the first vertebra and the ascending facet of the second vertebra; a first drill guide angled relative to the longitudinal axis of the central lumen, the first drill guide being configured to receive a drill therein so as to drill a first seat in the descending facet of the first vertebra; a second drill guide angled relative to the longitudinal axis of the central lumen, the second drill guide being configured to receive a drill therein so as to drill a second seat in the ascending facet of the second vertebra; wherein the first seat in the descending facet of the first vertebra is sized and angled so as to receive the superior stabilizer of the spinal facet fusion implant when the spinal facet fusion implant is advanced into the gap between the first vertebra and the second vertebra; and wherein the second seat in the ascending facet of the second vertebra is sized and angled so as to receive the inferior stabilizer of the spinal facet fusion implant when the spinal facet fusion implant is advanced into the gap between the first vertebra and the second vertebra.

(13) In another preferred form of the invention, there is provided a system for effecting spinal facet fusion, the system comprising: a spinal facet fusion implant, the spinal facet fusion implant comprising a taper; and a drill guide/cannula for preparing the anatomy to receive the spinal facet fusion implant and for delivering the spinal facet fusion implant to the anatomy, the drill guide/cannula comprising: a body having a distal end, a proximal end and a longitudinal axis

extending therebetween, the distal end of the body being configured for engaging the gap between a descending facet of a first vertebra and an ascending facet of a second vertebra; a central lumen extending between the distal end of the body and the proximal end of the body, the central lumen having a cross-sectional profile which matches the cross-sectional profile of the spinal facet fusion implant such that the spinal facet fusion implant can be introduced into the proximal end of the central lumen, advanced distally along the central lumen, and advanced distally out of the distal end of the central lumen and into the gap between the descending facet of the first vertebra and the ascending facet of the second vertebra; a first drill guide angled relative to the longitudinal axis of the central lumen, the first drill guide being configured to receive a drill therein so as to drill a first seat in the descending facet of the first vertebra; a second drill guide angled relative to the longitudinal axis of the central lumen, the second drill guide being configured to receive a drill therein so as to drill a second seat in the ascending facet of the second vertebra; wherein the first seat in the descending facet of the first vertebra and the second seat in the ascending facet of the second vertebra are sized and angled so as to receive the spinal facet fusion implant therein when the spinal facet fusion implant is advanced into the gap between the first vertebra and the second vertebra.

(14) In another preferred form of the invention, there is provided a method for effecting spinal facet fusion, the method comprising: providing a spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another; providing a drill guide/cannula for preparing the anatomy to receive the spinal facet fusion implant and for delivering the spinal facet fusion implant to the anatomy, the drill guide/cannula comprising: a body having a distal end, a proximal end and a longitudinal axis extending therebetween, the distal end of the body being configured for engaging the gap between a descending facet of a first vertebra and an ascending facet of a second vertebra; a central lumen extending between the distal end of the body and the proximal end of the body, the central lumen having a cross-sectional profile which matches the cross-sectional profile of the spinal facet fusion implant such that the spinal facet fusion implant can be introduced into the proximal end of the central lumen, advanced distally along the central lumen, and advanced distally out of the distal end of the central lumen and into the gap between the descending facet of the first vertebra and the ascending facet of the second vertebra; a first drill guide angled relative to the longitudinal axis of the central lumen, the first drill guide being configured to receive a drill therein; a second drill guide angled relative to the longitudinal axis of the central lumen, the second drill guide being configured to receive a drill therein; advancing a drill into the first drill guide so as to form a first seat in the descending facet of the first vertebra; advancing a drill into the second drill guide so as to form a second seat in the ascending facet of the second vertebra; advancing the spinal facet fusion implant into the gap between the first vertebra and the second vertebra so that (i) the superior stabilizer is disposed in the first seat of the descending facet of the first vertebra, and (ii) the inferior stabilizer is disposed in the second seat of the ascending facet of the second vertebra.

(15) In another preferred form of the invention, there is provided a method for effecting spinal facet fusion, the method comprising: providing a spinal facet fusion implant, the spinal facet fusion implant comprising a taper; forming a first seat in the descending facet of a first vertebra, and forming a second seat in the ascending facet of a second vertebra, wherein at least one of the first seat and the second seat comprises a taper; and advancing the spinal facet fusion implant into the

gap between the first vertebra and the second vertebra so that the spinal facet fusion implant is disposed in the first seat of the descending facet and the second seat of the ascending facet of the second vertebra.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other objects and features of the present invention will be more fully disclosed or rendered obvious by the following detailed description of the preferred embodiments of the invention, which is to be considered together with the accompanying drawings wherein like numbers refer to like parts, and further wherein:
- (2) FIGS. **1-3** are schematic views showing the seven cervical vertebrae (C1-C7);
- (3) FIG. **4** is a schematic view showing a facet joint between the C3 cervical vertebra and the C4 cervical vertebra;
- (4) FIGS. **5-10** are schematic views showing a novel fusion implant formed in accordance with the present invention;
- (5) FIGS. **11** and **12** are schematic views showing a joint locator which may be used to deploy the novel fusion implant of FIGS. **5-10**;
- (6) FIGS. **13** and **14** are schematic views showing a joint decorticator which may be used to decorticate bone to promote fusion;
- (7) FIGS. **15-17** are schematic views showing a drill guide/cannula which may be used to prepare bone to receive the novel fusion implant of FIGS. **5-10** and which may be used to deploy the novel fusion implant of FIGS. **5-10**;
- (8) FIGS. **18** and **19** are schematic views showing a tamp which may be used to deploy the novel fusion implant of FIGS. **5-10**;
- (9) FIGS. **20, 21, 21A, 22, 23, 23A, 24-26, 26A, 27-32, 32A, 33-36** are schematic views showing the novel fusion implant of FIGS. **5-10** being installed in a facet joint;
- (10) FIG. **36A** is a schematic view showing two novel fusion implants of FIGS. **5-10** installed in a pair of facet joints;
- (11) FIG. **37** is a schematic view showing how the novel fusion implant of FIGS. **5-10** interacts with the opposing facets of two adjacent cervical vertebrae so as to immobilize the facet joint and facilitate fusion between the opposing facets; and
- (12) FIGS. **38-42** are schematic views showing additional novel fusion implants formed in accordance with the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) The Cervical Vertebrae in General

(14) Looking first at FIGS. **1-3**, there is shown the seven cervical vertebrae (C1-C7) which are located at the top of the spine. FIGS. **1** and **2** show the seven cervical vertebrae being oriented vertically (e.g., such as when the patient is sitting or standing), and FIG. **3** shows the seven cervical vertebrae being oriented horizontally (e.g., such as when the patient is lying down). FIG. **4** shows an exemplary facet joint **5** located between a descending facet **10** of a C3 cervical vertebra **15** and an ascending facet **20** of a C4 cervical vertebra **25**. In facet joint **5**, face **30** of descending facet **10** of C3 cervical vertebra **15** engages face **35** of ascending facet **20** of C4 cervical vertebra **25**.

(15) The Novel Fusion Implant

(16) Looking next at FIGS. **5-10**, there is shown a novel spinal facet fusion implant **105** formed in accordance with the present invention. Fusion implant **105** generally comprises a body **110**, a superior stabilizer **115** and an inferior stabilizer **120**. As used herein, the term “superior stabilizer” is intended to identify the stabilizer which engages the descending facet of a facet joint (e.g., descending facet **10** of C3 cervical vertebra **15**), and the term “inferior stabilizer” is intended to

identify the stabilizer which engages the ascending facet of a facet joint (e.g., ascending facet **20** of a C4 cervical vertebra **25**).

(17) Body **110** comprises an elongated element having structural integrity. More particularly, body **110** generally comprises a distal end surface **125**, a proximal end surface **130**, a superior surface **135** extending distally from proximal end surface **130**, and an inferior surface **140** extending distally from proximal end surface **130**. Superior surface **135** and inferior surface **140** diverge as they extend distally from proximal end surface **130**, i.e., so that the “height” of the distal end of body **110** is greater than the “height” of the proximal end of body **110**. A superior beveled surface **145** connects the distal end of superior surface **135** with distal end surface **125**, and an inferior beveled surface **150** connects the distal end of inferior surface **140** with distal end surface **125**. Body **110** further comprises a medial side surface **155** and a lateral side surface **160**, each of which is bounded by the aforementioned distal end surface **125**, proximal end surface **130**, superior surface **135**, inferior surface **140**, superior beveled surface **145**, and inferior beveled surface **150**.

(18) In one preferred form of the invention, the “height” of proximal end surface **130** is sized to be approximately the width of the gap between the two facets of a facet joint.

(19) Superior stabilizer **115** generally comprises a distal end surface **165**, a proximal end surface **170**, a rounded superior surface **175** extending distally from proximal end surface **170**, and a superior beveled surface **180** which connects distal end surface **165** with generally rounded superior surface **175**. Superior stabilizer **115** further comprises a medial side surface **185** and a lateral side surface **190**, each of which is bounded by the aforementioned distal end surface **165**, proximal end surface **170**, rounded superior surface **175** and superior beveled surface **180**.

(20) Inferior stabilizer **120** generally comprises a distal end surface **195**, a proximal end surface **200**, a rounded inferior surface **205** extending distally from proximal end surface **200**, and an inferior beveled surface **210** which connects distal end surface **195** with generally rounded inferior surface **205**. Inferior stabilizer **120** further comprises a medial side surface **215** and a lateral side surface **220**, each of which is bounded by the aforementioned distal end surface **195**, proximal end surface **200**, rounded inferior surface **205** and inferior beveled surface **210**.

(21) In one preferred form of the invention, and as seen in FIGS. 5-10, proximal end surfaces **130**, **170** and **200** are substantially co-planar.

(22) In one preferred form of the invention, the tangent line **225** (FIG. 6) of rounded superior surface **175** of superior stabilizer **115** extends at an angle α (e.g., of approximately 10 degrees) relative to the longitudinal axis **230** of body **110**, and the tangent line **235** of rounded inferior surface **205** of inferior stabilizer **120** extends at an angle β (of approximately 15 degrees) relative to the longitudinal axis **230** of body **110**.

(23) In one preferred form of the invention, distal end surface **165** of superior stabilizer **115** is disposed distal to distal end surface **195** of inferior stabilizer **120**.

(24) Thus it will be seen that, in one preferred form of the invention, superior stabilizer **115** and inferior stabilizer **120** have different configurations.

(25) As will hereinafter be discussed in further detail, body **110** of novel fusion implant **105** is intended to be disposed the space between the descending facet of a facet joint and the opposing ascending facet of a facet joint (e.g., in the space between face **30** of descending facet **10** of C3 cervical vertebra **15** and face **35** of ascending facet **20** of C4 cervical vertebra **25**); superior stabilizer **115** is intended to be disposed in a seat formed in the descending facet of the facet joint (e.g., a seat formed in descending facet **10** of C3 cervical vertebra **15**); and inferior stabilizer **120** is intended to be disposed in a seat formed in the ascending facet of the facet joint (e.g., a seat formed in ascending facet **20** of C4 cervical vertebra **25**), whereby to immobilize the facet joint and facilitate fusing of the facet joint. In this respect it should be appreciated that, and as will hereinafter be discussed, fusion implant **105** preferably makes a “friction fit” with the two facets of the facet joint, and superior stabilizer **115** and inferior stabilizer **120** are seated in descending facet **10** and ascending facet **20**, respectively, so as to lock the facets against movement relative to one

another. Significantly, the tapered configuration of novel fusion implant **105**, when seated in an appropriately-configured pocket formed in the descending facet **10** and ascending facet **20** of a facet joint, locks the fusion implant against anterior or posterior movement. The tapered configuration of novel fusion implant **105** also helps create/restore lordosis.

(26) It should be appreciated that novel fusion implant **105** is particularly well suited for use in fusing a cervical facet joint.

(27) And as will hereinafter be discussed in further detail, novel fusion implant **105** is intended to be inserted into a facet joint using a posterior approach. Such a posterior approach is familiar to spine surgeons (thereby providing an increased level of comfort for the surgeon), and also minimizes the possibility of damage to the spinal cord during fusion implant insertion.

(28) Preferred Instrumentation for Installing the Novel Fusion Implant

(29) A preferred method for installing novel fusion implant **105** will hereinafter be described. The preferred method for installing novel fusion implant **105** preferably utilizes a joint locator **305** (FIGS. **11** and **12**), a joint decorticator **405** (FIGS. **13** and **14**), a drill guide/cannula **505** (FIGS. **15-17**), and a tamp **605** (FIGS. **18** and **19**).

(30) More particularly, joint locator **305** (FIGS. **11** and **12**) generally comprises a shaft **310** having a distal end **315** and a proximal end **320**. A finger **325** extends distally from distal end **315** of shaft **310**.

(31) Joint decorticator **405** (FIGS. **13** and **14**) generally comprises a hollow tubular structure **410** having a distal end **415** and a proximal end **420**. A cutting element **425** is disposed at distal end **415** of hollow tubular structure **410**, and a handle **430** is disposed at proximal end **420** of hollow tubular structure **410**.

(32) Drill guide/cannula **505** (FIGS. **15-17**) generally comprises a body **510** having a distal end **515** and a proximal end **520**. A central lumen **525** extends from distal end **515** to proximal end **520**. Central lumen **525** has a cross-sectional profile generally matching a longitudinal projection of the maximum profile of novel fusion implant **105**, such that fusion implant **105** can be advanced along the length of central lumen **525** in a controlled sliding motion. More particularly, the cross-sectional profile of central lumen **525** comprises a generally rectangular portion **530** for accommodating body **110** of fusion implant **105**, a first generally hemispherical portion **535** for accommodating superior stabilizer **115** of fusion implant **105**, and a second generally hemispherical portion **540** for accommodating inferior stabilizer **120** of fusion implant **105**.

(33) Drill guide/cannula **505** also comprises a first drill guide lumen **545**. First drill guide lumen **545** is oriented at an angle to the longitudinal axis of central lumen **525**. Note that the angle at which first drill guide lumen **545** is oriented relative to the longitudinal axis of central lumen **525** is the same angle β at which tangent line **235** of inferior stabilizer **120** extends to longitudinal axis **230** of body **110** of fusion implant **105**, such that first drill guide lumen **545** can be used to prepare a seat in the ascending facet of a facet joint (e.g., ascending facet **20** of C4 cervical vertebra **25**), as will hereinafter be discussed in further detail.

(34) Drill guide/cannula **505** also comprises a second drill guide lumen **550**. Second drill guide lumen **550** is oriented at an angle to the longitudinal axis of central lumen **525**. Note that the angle at which second drill guide lumen **550** is oriented relative to the longitudinal axis of central lumen **525** is the same angle α at which tangent line **225** of superior stabilizer **115** extends to longitudinal axis **230** of body **110** of fusion implant **105**, such that second drill guide lumen **550** can be used to prepare a seat in the descending facet of a facet joint (e.g., descending facet **10** of C3 cervical vertebra **15**), as will hereinafter be discussed in further detail.

(35) Drill guide/cannula **505** also comprises a pair of fingers **555** extending distally from its distal end **515**. Note that fingers **555** are aligned in a first plane **560** (FIG. **15**) which extends perpendicular to a second plane **565** (FIG. **17**) which passes through first drill guide lumen **545** and second drill guide lumen **550** so that, when fingers **555** are inserted into the gap in the facet joint (see below), first drill guide lumen **545** and second drill guide lumen **550** may be used to prepare

seats in the descending facet of a facet joint, and the ascending facet of a facet joint, respectively (as will hereinafter be discussed). Note also that the aforementioned second plane **565** bisects first generally hemispherical portion **535** of central lumen **525** and second generally hemispherical portion **540** of central lumen **525** so that, when fingers **555** are inserted into the gap in the facet joint (see below), first generally hemispherical portion **535** and second generally hemispherical portion **540** may be used to orient fusion implant **105** relative to the seats previously prepared in the descending facet of a facet joint, and the ascending facet of a facet joint, respectively (as will hereinafter be discussed).

(36) Tamp **605** (FIGS. **18** and **19**) comprises a shaft **610** having a distal end **615** and a proximal end **620**. Shaft **610** preferably has a rectangular cross-section similar to generally rectangular portion **530** of central lumen **525** of drill guide/cannula **505**. A finger **625** extends distally from distal end **615** of shaft **610**. An impactor extension **630** is mountable to proximal end **620** of shaft **610**, such that hammering on the proximal end of impactor extension **630** will transmit distally-directed force to shaft **610**. A superior projection **635** is disposed on shaft **610** just proximal to finger **625**. Superior projection **635** has a hemispherical cross-section similar to first generally hemispherical portion **535** of central lumen **525** of drill guide/cannula **505**. As a result of this construction, tamp **605** can make a close sliding fit within central lumen **525** of drill guide/cannula **505**, whereby to enable tamp **605** to advance fusion implant **105** along the length of central lumen **525**, as will hereinafter be discussed. In addition, when impactor extension **630** is mounted to proximal end **620** of shaft **610**, hammering on the proximal end of impactor extension **630** will transmit distally-directed force to shaft **610** of tamp **605**, whereby to transmit distally-directed force to fusion implant **105**, as will hereinafter be discussed.

(37) Preferred Method for Installing the Novel Fusion Implant

(38) First, the facet joint is visualized by the surgeon, either indirectly by imaging the patient (e.g., by fluoroscopy) or directly by visualization during an open procedure.

(39) Next, and looking now at FIGS. **20**, **21** and **21A**, joint locator **305** is inserted into the gap between the opposing facet surfaces (e.g., between face **30** of descending facet **10** of C3 cervical vertebrae **15** and face **35** of ascending facet **20** of C4 cervical vertebrae **25**). More particularly, finger **325** of joint locator **305** is advanced into the gap between the opposing facet surfaces and then the position of joint locator **305** is verified, e.g., by viewing along the coronal and sagittal planes of the patient. After verification is complete, the proximal end of joint locator **305** is lightly tapped so as to advance finger **325** of joint locator **305** further into the facet joint, until distal end **315** of shaft **310** engages the outer surfaces of the facets. At this point, joint locator **305** is essentially fixed to the facet joint and provides a “track” to the facet joint.

(40) Next, and looking now at FIGS. **22**, **23** and **23A**, joint decorticator **405** is slid over joint locator **305** so that the joint decorticator is aligned with the gap between the opposing facet surfaces (e.g., between face **30** of descending facet **10** of C3 cervical vertebrae **15** and face **35** of ascending facet **20** of C4 cervical vertebrae **25**). Then joint decorticator **405** is used to decorticate the ends of the opposing facets. After decortication is completed, joint decorticator **405** is removed from joint locator **305**.

(41) Then, and looking now at FIG. **24**, drill guide/cannula **505** is advanced over joint locator **305** (i.e., by fitting the distal end of central lumen **525** of drill guide/cannula **505** over proximal end **320** of joint locator **305**, and then advancing the drill guide/cannula distally along joint locator **305**). Drill guide/cannula **505** is advanced distally along joint locator **305** until fingers **555** of drill guide/cannula **505** enter the gap between the opposing facets. Then fingers **555** of drill guide/cannula **505** are driven into the gap between the opposing facets, whereby to secure drill guide/cannula **505** to the facet joint.

(42) Next, and looking now at FIGS. **25**, **26** and **26A**, joint locator **305** is removed, e.g., by pulling the joint locator proximally out of central lumen **525** of the facet joint and out of drill guide/cannula **505**.

(43) At this point, and looking now at FIGS. 27 and 28, a drill 705 is inserted into first drill guide lumen 545 of drill guide/cannula 505 and advanced into the ascending facet of the facet joint (e.g., ascending facet 20 of C4 cervical vertebrae 25) so as to create a recess 710 in the ascending facet of the facet joint. Recess 710 will subsequently receive inferior stabilizer 120 of fusion implant 105, as will hereinafter be discussed. Then drill 705 is removed from first drill guide lumen 545.

(44) Next, and looking now at FIGS. 29 and 30, a drill 715 (which may or may not be the same as the aforementioned drill 705) is inserted into second drill guide lumen 550 of drill guide/cannula 505 and advanced into the descending facet of the facet joint (e.g., descending facet 10 of C3 cervical vertebrae 15) so as to create a recess 720 in the descending facet of the facet joint. Recess 720 will subsequently receive superior stabilizer 115 of fusion implant 105, as will hereinafter be discussed. Then drill 715 is removed from second drill guide lumen 550.

(45) At this point, and looking now at FIG. 31, a pocket 725 will have been created in the facet joint, i.e., by forming recess 710 in the ascending facet of the facet joint and forming recess 720 in the descending facet of the facet joint. Note that pocket 725 is generally characterized by a superior anterior stop surface 730, an inferior anterior stop surface 735, a superior posterior stop surface 740 and an inferior posterior stop surface 745.

(46) Next, and looking now at FIGS. 32-36, fusion implant 105 is installed into the facet joint using drill guide/cannula 505 and tamp 605.

(47) More particularly, fusion implant 105 is inserted into the proximal end of central lumen 525 of drill guide/cannula 505 (FIGS. 31 and 32) so that body 110 of fusion implant 105 is received in generally rectangular portion 530 of central lumen 525, superior stabilizer 115 of fusion implant 105 is received in first generally hemispherical portion 535 of central lumen 525, and inferior stabilizer 120 is received in second generally hemispherical portion 540 of central lumen 525.

(48) Then tamp 605 is used to advance fusion implant 105 along central lumen 525 of drill guide/cannula 505, and then tamp 605 and impactor extension 630 are used to hammer fusion implant 105 into the aforementioned pocket 725 formed in the facet joint (FIGS. 33-36), with body 110 of fusion implant 105 spanning the gap between the descending facet of the facet joint and the ascending facet of the facet joint, and with superior stabilizer 115 of fusion implant 105 being received in recess 720 in the descending facet of the facet joint, and with inferior stabilizer 120 of fusion implant 105 being received in recess 710 in the ascending facet of the facet joint, whereby to immobilize the facet joint and facilitate fusing of the facet joint. After fusion implant 105 is installed in pocket 725 in the facet joint, tamp 605 and drill guide/cannula 505 are removed from the surgical site, thereby completing the installation of fusion implant 105 into the facet joint.

(49) FIG. 32A shows two fusion implants 105 disposed in two facet joints in a spine.

(50) Significantly, and looking now at FIG. 37, inasmuch as fusion implant 105 is configured to be “taller” at its distal end than at its proximal end, and inasmuch as fusion implant 105 is configured to have a “height” at its proximal end which is approximately the same as the “height” of the gap between the two facets of the facet joint, the two facets are distracted somewhat during insertion of the fusion implant into the aforementioned pocket 725. As fusion implant 105 settles into pocket 725, the facets are able to return toward their undistracted condition and compress back against the fusion implant, essentially creating something of a “friction fit” between fusion implant 105 and the two facets of facet joint 5. By forming fusion implant 105 with angled surfaces, and by forming the aforementioned pocket 725 in the facets with angled surfaces, the fusion implant is prevented from moving anteriorly or posteriorly once the facets have compressed back into their normal disposition. More particularly, movement of fusion implant 105 in the anterior direction is prevented by engagement of superior stabilizer 115 and inferior stabilizer 120 with superior anterior stop surface 730 of pocket 725 and inferior anterior stop surface 735 of pocket 725, respectively. Movement of fusion implant 105 in the posterior direction is prevented by engagement of superior stabilizer 115 and inferior stabilizer 120 with superior posterior stop surface 740 of pocket 725 and inferior posterior stop surface 745 of pocket 725, respectively. Thus

it will be appreciated that, in order for superior stabilizer **115** and inferior stabilizer **120** to pass by superior posterior stop surface **740** and inferior posterior stop surface **745**, respectively, the facet joint would have to over-distract, which is inhibited by the surrounding soft tissue structure.

(51) By way of example but not limitation, where fusion implant **105** is to be seated between the C3 cervical vertebra and the C4 cervical vertebra (FIG. 37), as fusion implant **105** is advanced into the gap between face **30** of descending facet **10** of C3 cervical vertebra **15** and face **35** of ascending facet **20** of C4 cervical vertebra **25**, superior beveled surface **145** of fusion implant **105** and inferior beveled surface **150** of fusion implant **105** will engage descending facet **10** of C3 cervical vertebra **15** and ascending facet **20** of C4 cervical vertebra **25**, respectively, and cam the two facets apart. Further insertion of fusion implant **105** into the facet joint causes superior beveled surface **180** of superior stabilizer **115** and inferior beveled surface **210** of inferior stabilizer **120** to engage descending facet **10** of C3 cervical vertebra **15** and ascending facet **20** of C4 cervical vertebra **25**, respectively, and cam the two facets further apart. As fusion implant **105** settles into pocket **725**, face **30** of descending facet **10** of C3 cervical vertebra **15** settles against superior surface **135** of body **110**, face **35** of ascending facet **20** of C4 cervical vertebra **25** settles against inferior surface **140** of body **110**, superior anterior stop surface **730** of pocket **725** settles against distal end surface **165** of superior stabilizer **115**, inferior anterior stop surface **735** of pocket **725** settles against distal end surface **195** of inferior stabilizer **120**, superior posterior stop surface **740** of pocket **725** settles against rounded superior surface **175** of superior stabilizer **115** and inferior posterior stop surface **745** of pocket **725** settles against rounded inferior surface **205** of inferior stabilizer **120**, whereby to lock fusion implant **105** against anterior or posterior movement.

(52) Additionally, the wedge-like construction of fusion implant **105** creates/restores lordosis.

(53) In the preferred form of the invention, descending facet **10** contacts, and is supported by, superior beveled surface **145**, superior surface **135**, distal end surface **165**, superior beveled surface **180** and rounded superior surface **175** of fusion implant **105**; and ascending facet **20** contacts, and is supported by, inferior beveled surface **150**, inferior surface **140**, distal end surface **195**, inferior beveled surface **210** and rounded inferior surface **205** of fusion implant **105**.

(54) However, if desired, in another form of the invention, descending facet **10** contacts, and is supported by, less than all of the aforementioned superior beveled surface **145**, superior surface **135**, distal end surface **165**, superior beveled surface **180** and rounded superior surface **175** of fusion implant **105**; and ascending facet **20** contacts, and is supported by, less than all of the aforementioned inferior beveled surface **150**, inferior surface **140**, distal end surface **195**, inferior beveled surface **210** and rounded inferior surface **205** of fusion implant **105**. By way of example but not limitation, in another form of the invention, descending facet **10** contacts, and is supported by, only superior surface **135** of fusion implant **105** and ascending facet **20** contacts, and is supported by, only inferior surface **140** of fusion implant **105**. By way of further example but not limitation, in another form of the invention, descending facet **10** contacts, and is supported by, only rounded superior surface **175** of fusion implant **105** and ascending facet **20** contacts, and is supported by, only rounded inferior surface **205** of fusion implant **105**.

(55) Alternative Constructions

(56) If desired, fusion implant **105** may comprise visual markers to facilitate proper orientation of the fusion implant when fusion implant **105** is inserted into the proximal end of central lumen **525** of drill guide/cannula **505**. By way of example but not limitation, and looking now at FIG. 38, fusion implant **105** may comprise a black dot **750** on one or both of inferior stabilizer **120** (e.g., to identify the inferior side of fusion implant **105**) and/or distal end surface **125** of body **110** (e.g., to identify the distal end of fusion implant **105**).

(57) The configuration of fusion implant **105** may be varied without departing from the scope of the present invention.

(58) By way of example but not limitation, it should be appreciated that the new fusion implant may be manufactured in a wide range of different sizes in order to accommodate any size of facet

joint.

(59) Furthermore, the scale and aspect ratio of body **110**, superior stabilizer **115** and inferior stabilizer **120** may all be varied without departing from the scope of the present invention.

(60) See, for example, FIG. **39**, which shows a fusion implant **105** wherein the configurations of superior stabilizer **115** and inferior stabilizer **120** differ from the configurations of superior stabilizer **115** and inferior stabilizer **120** in the fusion implant **105** shown in FIGS. **5-10**.

(61) By way of further example but not limitation, fusion implant **105** may be formed so that superior stabilizer **115** and inferior stabilizer **120** have identical configurations.

(62) See, for example, FIG. **40**, where superior stabilizer **115** and inferior stabilizer **120** have identical configurations (and where those configurations vary from the configurations of superior stabilizer **115** and inferior stabilizer **120** in the fusion implant **105** shown in FIGS. **5-10**).

(63) See also, for example, FIG. **41**, where the apexes of rounded superior surface **175** of superior stabilizer **115** and rounded inferior surface **205** of inferior stabilizer **120** extend substantially parallel to one another.

(64) And see, for example, FIG. **42**, where superior surface **135** and inferior surface **140** of body **110** extend substantially parallel to one another.

(65) It will, of course, be appreciated that the configuration of drill guide/cannula **505** may change as the configuration of fusion implant **105** changes in order to (i) enable drill guide/cannula **505** to create an appropriate pocket **725** in the facet joint, and (ii) enable drill guide/cannula **505** to provide an appropriate channel for advancing fusion implant **105** into pocket **725** in the facet joint.

(66) Additionally, the new fusion implant may be constructed out of any substantially biocompatible material which has properties consistent with the present invention including, but not limited to, allograft, autograft, synthetic bone, simulated bone material, biocomposites, ceramics, PEEK, stainless steel and titanium. Thus, the present invention permits the surgeon to select a fusion implant having the appropriate size and composition for a given facet fusion.

Advantages of the Invention

(67) Numerous advantages are achieved by the present invention. Among other things, the present invention provides a fast, simple, minimally-invasive, easily reproduced and highly effective approach for effecting facet fusion, particularly with cervical facet joints. The fusion implant is able to withstand greater forces, prohibit motion in all directions and drastically reduce the risk of implant failure. The fusion implant also eliminates the possibility of slippage during spinal motion, greatly improves facet stability and promotes better facet fusion.

(68) Applications to Joints Other than Facet Joints

(69) While fusion implant **105** has been discussed above in the context of fusing a facet joint, it should also be appreciated that fusion implant **105** may be used to stabilize and fuse any joint having anatomy similar to the facet joint, i.e., a pair of opposing bony surfaces defining a gap therebetween, with the stabilizer of the fusion implant being sized to be positioned within the gap. By way of example but not limitation, the fusion implant may be used in small joints such as the fingers, toes, etc.

Modifications of the Preferred Embodiments

(70) It should be understood that many additional changes in the details, materials, steps and arrangements of parts, which have been herein described and illustrated in order to explain the nature of the present invention, may be made by those skilled in the art while still remaining within the principles and scope of the invention.

Claims

1. A spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer

extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another; wherein the elongated body comprises a proximal end surface, the superior stabilizer comprises a proximal end surface, and the inferior stabilizer comprises a proximal end surface, and further wherein the proximal end surface of the elongated body, the proximal end surface of the superior stabilizer, and the proximal end surface of the inferior stabilizer are all co-planar.

2. A spinal facet fusion implant according to claim 1 wherein the superior body surface and the inferior body surface are tapered relative to one another.

3. A spinal facet fusion implant according to claim 1 wherein the superior body surface and the inferior body surface are tapered inwardly relative to one another in a proximal direction.

4. A spinal facet fusion implant according to claim 3 wherein the superior stabilizer surface and the inferior stabilizer surface extend parallel to one another.

5. A spinal facet fusion implant according to claim 1 wherein the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another.

6. A spinal facet fusion implant according to claim 1 wherein the superior stabilizer surface and the inferior stabilizer surface are tapered inwardly relative to one another in a proximal direction.

7. A spinal facet fusion implant according to claim 6 wherein the superior body surface and the inferior body surface extend parallel to one another.

8. A spinal facet fusion implant according to claim 1 wherein the superior body surface and the inferior body surface are tapered relative to one another, and wherein the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another.

9. A spinal facet fusion implant according to claim 8 wherein the superior body surface and the inferior body surface are tapered inwardly relative to one another in the proximal direction, and wherein the superior stabilizer surface and the inferior stabilizer surface are tapered inwardly relative to one another in a proximal direction.

10. A spinal facet fusion implant according to claim 1 wherein the spinal facet fusion implant is asymmetric about the longitudinal axis of the elongated body.

11. A spinal facet fusion implant according to claim 10 wherein the elongated body is symmetrical about the longitudinal axis of the elongated body.

12. A spinal facet fusion implant according to claim 11 wherein the superior stabilizer and the inferior stabilizer have differing profiles in side view.

13. A spinal facet fusion implant according to claim 1 wherein the spinal facet fusion implant is symmetrical about the longitudinal axis of the elongated body.

14. A spinal facet fusion implant according to claim 1 wherein the distal end of the elongated body terminates in a distal end surface, wherein the elongated body comprises a beveled surface extending between the superior body surface and the distal end surface, and wherein the elongated body comprises a beveled surface extending between the inferior body surface and the distal end surface.

15. A spinal facet fusion implant according to claim 12 wherein the beveled surface extending between the superior body surface and the distal end surface extends at a first angle to the longitudinal axis of the elongated body, and the beveled surface extending between the inferior body surface and the distal end surface extends at a second angle to the longitudinal axis of the elongated body, and further wherein the first angle and the second angle are of the same size.

16. A spinal facet fusion implant according to claim 1 wherein the superior stabilizer comprises a beveled surface extending between the superior stabilizer and the superior body surface, and wherein the inferior stabilizer comprises a beveled surface extending between the inferior stabilizer and the inferior body surface.

17. A spinal facet fusion implant according to claim 16 wherein the beveled surface extending between the superior stabilizer and the superior body surface is disposed distal to the beveled surface extending between the inferior stabilizer and the inferior body surface.

18. A spinal facet fusion implant according to claim 1 wherein the superior stabilizer comprises two beveled surfaces extending between the superior stabilizer and the superior body surface, and wherein the inferior stabilizer comprises two beveled surfaces extending between the inferior stabilizer and the at least one inferior body surface.

19. A spinal facet fusion implant according to claim 18 wherein the two beveled surfaces extending between the superior stabilizer and the superior body surface extend at different angles to the longitudinal axis of the elongated body.

20. A spinal facet fusion implant according to claim 18 wherein the two beveled surfaces extending between the inferior stabilizer and the inferior body surface extend at different angles to the longitudinal axis of the elongated body.

21. A system for effecting spinal facet fusion, the system comprising: a spinal facet fusion implant comprising: an elongated body having a distal end, a proximal end and a longitudinal axis extending between the distal end and the proximal end, the elongated body being characterized by a superior body surface and an inferior body surface; a superior stabilizer extending outwardly from the superior body surface, the superior stabilizer being characterized by a superior stabilizer surface; and an inferior stabilizer extending outwardly from the inferior body surface, the inferior stabilizer being characterized by an inferior stabilizer surface; wherein (i) the superior body surface and the inferior body surface are tapered relative to one another, and/or (ii) the superior stabilizer surface and the inferior stabilizer surface are tapered relative to one another; wherein the elongated body comprises a proximal end surface, the superior stabilizer comprises a proximal end surface, and the inferior stabilizer comprises a proximal end surface, and further wherein the proximal end surface of the elongated body, the proximal end surface of the superior stabilizer, and the proximal end surface of the inferior stabilizer are all co-planar; a drill guide/cannula for preparing the anatomy to receive the spinal facet fusion implant and for delivering the spinal facet fusion implant to the anatomy, the drill guide/cannula comprising: a body having a distal end, a proximal end and a longitudinal axis extending therebetween, the distal end of the body being configured for engaging a gap between a descending facet of a first vertebra and an ascending facet of a second vertebra; a central lumen extending between the distal end of the body and the proximal end of the body, the central lumen having a cross-sectional profile which matches the cross-sectional profile of the spinal facet fusion implant such that the spinal facet fusion implant can be introduced into the proximal end of the central lumen, advanced distally along the central lumen, and advanced distally out of the distal end of the central lumen and into the gap between the descending facet of the first vertebra and the ascending facet of the second vertebra; a first drill guide angled relative to the longitudinal axis of the central lumen, the first drill guide being configured to receive a drill therein so as to drill a first seat in the descending facet of the first vertebra; a second drill guide angled relative to the longitudinal axis of the central lumen, the second drill guide being configured to receive a drill therein so as to drill a second seat in the ascending facet of the second vertebra; wherein the first seat in the descending facet of the first vertebra is sized and angled so as to receive the superior stabilizer of the spinal facet fusion implant when the spinal facet fusion implant is advanced into the gap between the first vertebra and the second vertebra; and wherein the second seat in the ascending facet of the second vertebra is sized and angled so as to receive the inferior stabilizer of the spinal facet fusion implant when the spinal facet fusion implant is advanced into the gap between the first vertebra and the second vertebra.
